C++ DSA Pattern Practice

10 combined Easy, Medium & Hard pattern problems to strengthen logic, loops, nested loops, spaces, symmetry, and pattern construction.

Level	Questions	Focus
Easy	1–3	Nested loops, stars, numbers, basic spaces
Medium	4–7	Symmetry, pyramids, hollow patterns, mixed logic
Hard	8–10	Complex symmetry, palindromic structure, advanced spacing

Practice rule: First solve using only **for/while loops, if/else, and variables**. Avoid copying pattern code. For each problem, identify the number of rows, spaces, and symbols printed in every row before writing the C++ code.

Easy — Q1. Increasing Star Triangle

Problem: Given an integer N, print a left-aligned triangle of stars.

Example (N = 5):

```
*
**
***
****
*****
```

> **Hint:** Think about how the number of stars changes from row to row.

Your solution: _____

Easy — Q2. Number Triangle

Problem: Given N, print a triangle where row i contains the number i repeated i times.

Example (N = 5):

```
1
22
333
4444
55555
```

> **Hint:** Use the outer loop for rows and decide what the inner loop should print.

Your solution: _____

Easy — Q3. Reverse Star Triangle

Problem: Given N, print a left-aligned inverted triangle of stars.

Example (N = 5):

```
*****
****
***
```

**
*

Hint: Compare the number of stars in each row with the row number.

Your solution: _____

Medium — Q4. Right-Aligned Star Triangle

Problem: Given N, print a right-aligned triangle of stars. The output should contain leading spaces.

Example (N = 5):

```
    *
   **
  ***
 ****
*****
```

Hint: For every row, calculate both spaces and stars. Do not hard-code spaces.

Your solution: _____

Medium — Q5. Full Pyramid

Problem: Given N, print a centered pyramid. Each row should contain spaces followed by an odd number of stars.

Example (N = 5):

```
    *
   ***
  *****
 *******
*********
```

Hint: Find the formula for spaces and stars using the row number.

Your solution: _____

Medium — Q6. Hollow Square

Problem: Given N, print a hollow square of stars. Only the border should contain stars.

Example (N = 5):

```
*****
*...*
*...*
*...*
*****
```

Hint: Inside the square, print '*' only when the cell is on a border.

Your solution: _____

Medium — Q7. Diamond Pattern

Problem: Given N, print a symmetric diamond made of stars. N represents the number of rows in the upper half.

Example (N = 4):

```
   *
  ***
 *****
*******
 *****
  ***
   *
```

Hint: Solve the upper pyramid first, then mirror the logic for the lower half.

Your solution: _____

Hard — Q8. Palindromic Number Pyramid

Problem: Given N, print a centered pyramid of numbers where each row increases toward the center and then decreases.

Example (N = 5):

```
    1
   121
  12321
 1234321
123454321
```

Hint: Separate the row into spaces, increasing numbers, and decreasing numbers.

Your solution: _____

Hard — Q9. Hollow Diamond

Problem: Given N, print a hollow diamond. Only the boundary positions should contain stars; the inside must remain empty.

Example (N = 4):

```
   *
  * *
```

```
 *  *
*   *
 *  *
  * *
   *
```

Hint: This combines diamond symmetry with a border condition. Carefully derive the first and last star positions.

Your solution: _____

Hard — Q10. Butterfly Pattern

Problem: Given N, print a butterfly pattern using stars. The top half grows from N stars to 2N stars with a gap; the bottom half mirrors it.

Example (N = 4):

```
****    ****
***      ***
**        **
*          *
**        **
***      ***
****    ****
```

Hint: Think in three parts per row: left stars, middle spaces, right stars. Then mirror the row count.

Your solution: _____

Pattern-Solving Checklist

1. Identify the total number of rows.
2. For each row, determine the number of spaces.
3. Determine the number of stars/numbers/symbols.
4. Find the relationship between row number and printed elements.
5. Decide whether the pattern can be split into upper and lower halves.
6. Convert the logic into nested loops.
7. Use conditions only where the shape requires them (for example, hollow borders).
8. Test with small values such as N = 1, 2, 3, and 5.
9. After solving, explain your code line-by-line without looking at it.
10. Try rewriting the same pattern using while loops after completing the for-loop version.

Recommended order: Q1 → Q2 → Q3 → Q4 → Q5 → Q6 → Q7 → Q8 → Q9 → Q10. Do not move to the next level until you can explain the row/space/symbol logic of the previous pattern.